



PowerPoint Presentation by
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MANAGEMENT ACCOUNTING

8th EDITION

BY

HANSEN & MOWEN

9 STANDARD COSTING

LEARNING OBJECTIVES

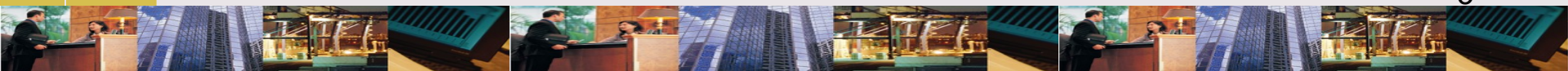
After studying this chapter, you should be able to:



LEARNING OBJECTIVES

1. Tell how unit standards are set; why standard costing systems are adopted.
2. State the purpose of a standard cost sheet.
3. Describe basic concepts underlying variance analysis & explain how they are used for control.

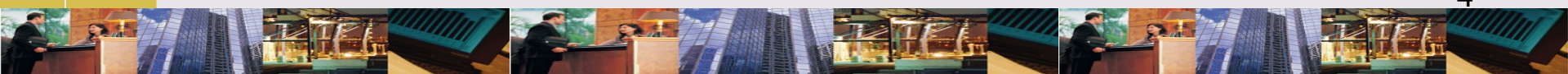
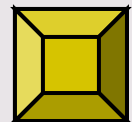
Continued



LEARNING OBJECTIVES

4. Compute materials & labor variances; explain how they are used for control.
5. Calculate variable & fixed overhead variances & give their definitions.
6. Prepare journal entries for variances (*Appendix*).

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT:

Blue Corn Foods

What motivated Rosita to implement a more formal cost control system?



QUESTIONS TO THINK ABOUT:

Blue Corn Foods

Why does a standard cost system provide more detailed control information?



QUESTIONS TO THINK ABOUT:

Blue Corn Foods

What type of control is being exercised with the use of standards?



QUESTIONS TO THINK ABOUT:

Blue Corn Foods

How can standards be used to control costs?



LEARNING OBJECTIVE

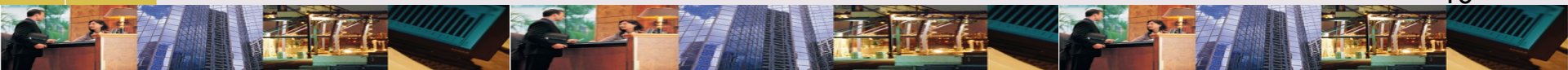
1

Tell how unit standards are set; why standard costing systems are adopted.

QUANTITY STANDARDS:

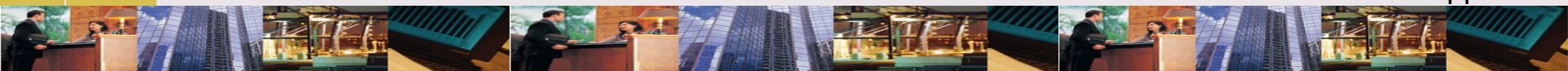
Definition

Tell the amount of input that **should** be used per unit of output.



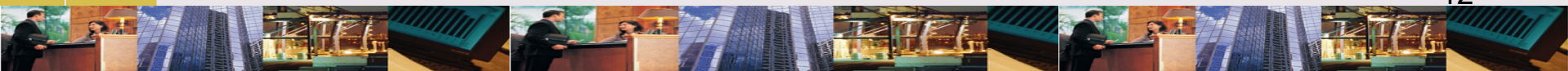
PRICE STANDARDS: Definition

Tell the amount that should be paid for the quantity of input used.



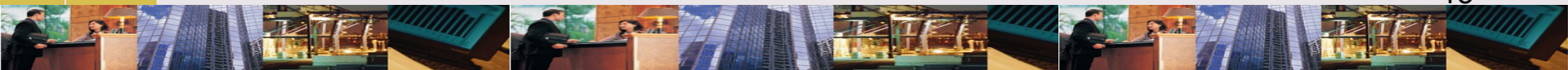
Where do quantity & price standards come from?

Quantity standards come from experience, studies, & personnel.
Price standards come from operations, purchasing, personnel, & accounting.



What is the difference
between ideal and
attainable standards?

Ideal standards only work under perfect conditions. **Attainable standards** can be achieved under efficient operating conditions.



STANDARD COST SYSTEMS

Why adopt a standard cost system?

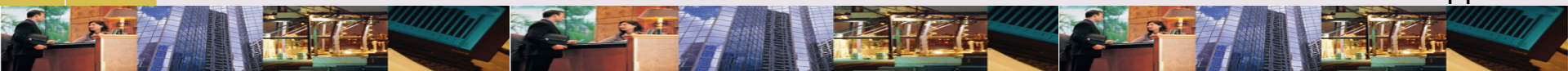
- ✧ For planning & control

 - ✧ To improve performance measures

 - ✧ To give manager more information by decomposing total variances into price & usage variances

- ✧ For product costing

 - ✧ To use unit cost system that is readily available in pricing



COST ASSIGNMENT

Standard costs are readily available for product costing in a standard cost system.

	Manufacturing Costs		
	Direct Materials	Direct Labor	Overhead
Actual costing system	Actual	Actual	Actual
Normal costing system	Actual	Actual	Budgeted
Standard costing system	Standard	Standard	Standard

Exhibit 9-1 Cost Assignment Approaches

EXHIBIT 9-1

LEARNING OBJECTIVE

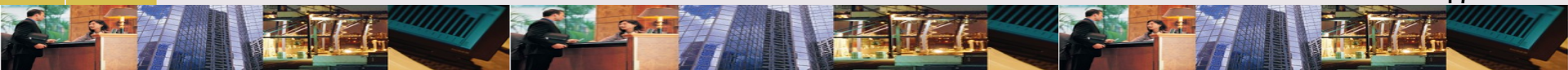
2

State the purpose of a standard cost sheet.

STANDARD COST PER UNIT:

Definition

Is the sum of standards costs for direct materials (DM), direct labor (DL), & overhead.



BLUECHITO COST SHEET

Standard cost sheet provides details for standard cost measures.

Description	Standard Price	Standard Usage	Standard Cost*	Subtotal
Direct materials:				
Blue corn	\$ 0.006	18 oz.	\$0.108	
Cooking oil	0.031	2 oz.	0.062	
Salt	0.005	1 oz.	0.005	
Lime	0.400	0.01 oz.	0.004	
Bags	0.044	1 bag	<u>0.044</u>	
Total direct materials				\$0.223
Direct labor:				
Inspectors	7.000	0.0070 hr.	\$0.049	
Machine operators	10.000	0.0008 hr.	<u>0.008</u>	
Total direct labor				0.057
Overhead:				
Variable overhead	3.850	0.0078 hr.	\$0.030	
Fixed overhead	32.050	0.0078 hr.	0.250**	
Total overhead				<u>0.280</u>
Total standard unit cost				<u>\$0.560</u>

*Calculated by multiplying price times usage.
**Rounded

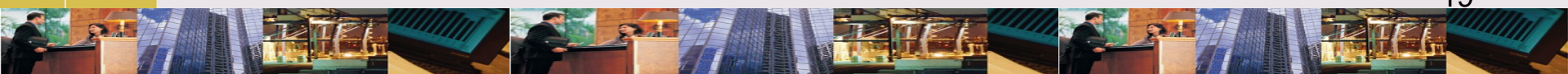
Exhibit 9-2 Standard Cost Sheet for Bluechitos

EXHIBIT 9-2

LEARNING OBJECTIVE

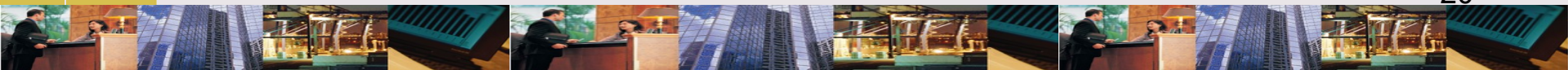
3

Describe basic concepts underlying variance analysis & explain how they are used for control.



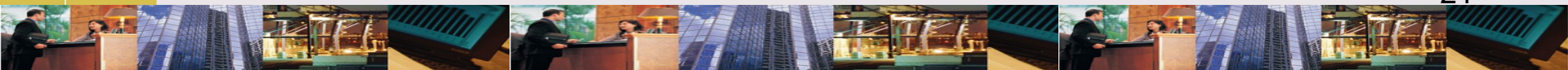
TOTAL BUDGET VARIANCE: Definition

Is the difference between **actual**
cost & **planned** cost of
production.



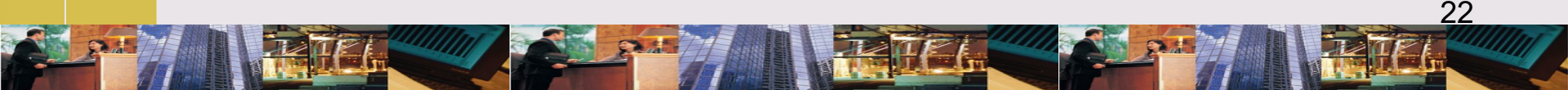
FAVORABLE & UNFAVORABLE

The difference between actual & planned can be **favorable** (actual price or usage $<$ standard) or **unfavorable** (actual price or usage $>$ standard).. Does not mean good or bad!!



What should we do when
we find variances?

If variances are significant, that is
if they are beyond our control
limits, they should be investigated
if it is cost beneficial to do so.



FORMULA: Total Variance

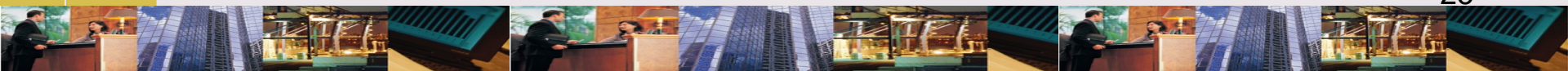
*Total variance is Actual cost – Applied cost or
Total cost – Standard cost.*

Total Variance

$$= (\text{AP} \times \text{AQ}) - (\text{SP} \times \text{SQ})$$

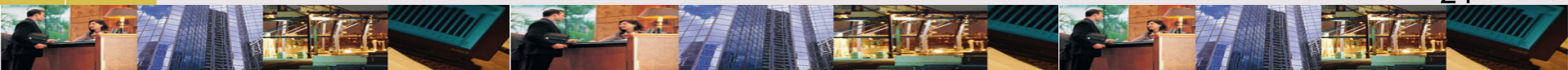
= Actual price x Actual quantity

– Standard Price x Standard Quantity



How can we make total variances more useful?

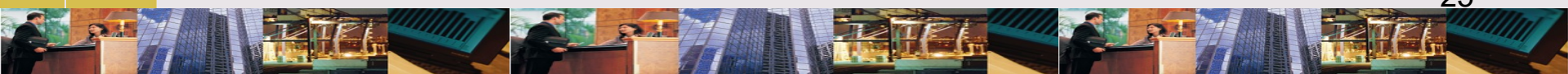
Total variances provide more information if they are divided into **Price** variances & **Efficiency** variances.



LEARNING OBJECTIVE

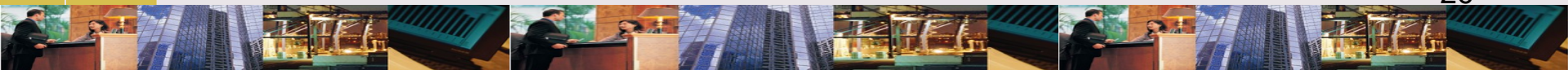
4

Compute materials & labor variances; explain how they are used for control.



FAVORABLE & UNFAVORABLE

Repeat: The difference between actual & planned can be **favorable** ((actual price or usage < standard)) or **unfavorable** ((actual price or usage > standard)). Does not mean good or bad!!



BLUE-CORN FOODS, INC.:

Background

Information for **actual** production, cost of corn, & inspectors.

Actual production	48,500 bags corn chips
Actual cost of corn	780,000 ounces @ \$0.0069
Actual cost of inspectors	360 hours @ \$7.35 = \$2,646



TOTAL MATERIALS & LABOR VARIANCES

Compares **actual costs** with **budgeted costs** at level of production.

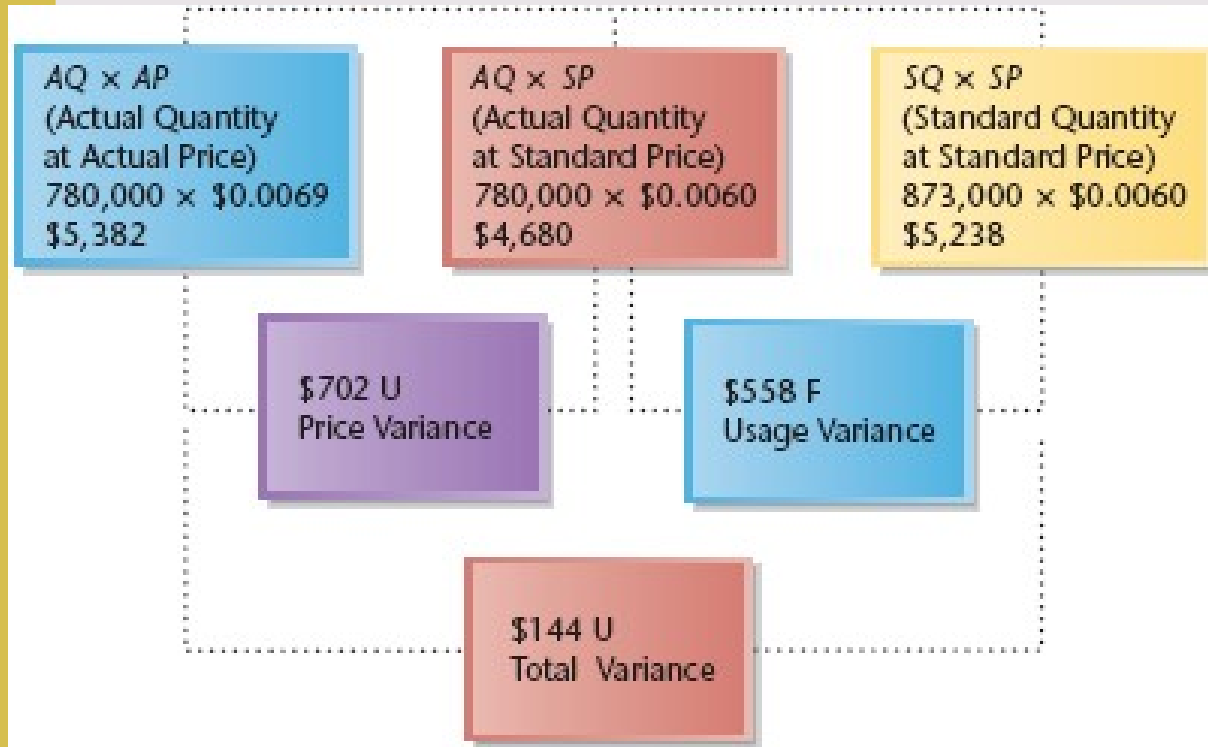
	Actual Costs	Budgeted Costs*	Total Variance
Corn	\$5,382.00	\$5,238.00	\$144.00 U
Inspectors labor	2,646.00	2,376.50	269.50 U

*The standard quantities for materials and labor are computed as unit quantity standards from Exhibit 9-2:
 Corn: $18 \times 48,500 = 873,000$ ounces
 Labor: $0.007 \times 48,500 = 339.5$ hours
 Multiplying these standard quantities by the unit standard prices given in Exhibit 9-2 produces the budgeted amounts appearing in this column:
 Corn: $\$0.006 \times 873,000 = \$5,238.00$
 Labor: $\$7.00 \times 339.5 = \$2,376.50$

Exhibit 9-5 Performance Report: Total Variances

EXHIBIT 9-5

MATERIALS VARIANCES



Decompose total materials variance into **price** & **usage** variances.

EXHIBIT 9-6

FORMULA: Materials Price Variance (MPV)

Materials price variance tells whether a company paid more than expected for raw materials.

MPV

$$= (AQ \times AP) - (AQ \times SP)$$

$$= (AP - SP)AQ$$

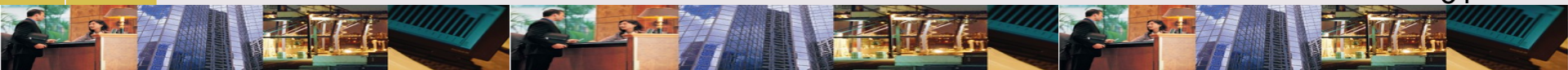
$$= (\$0.0069 - \$0.0060) 780,000$$

$$= \$ 702 \text{ U}$$



Who is responsible for a materials price variance?

The Purchasing Agent.



MPV AS PERFORMANCE EVALUATION

Limitations on using price variance in performance evaluation: buying lower quality or too much inventory. Results of investigation show shortage of usual grade of corn; purchasing agent has no control over supply.



FORMULA: Materials Usage Variance (MUV)

Materials usage variance tells whether a company used more raw materials than expected.

MUV

$$= (AQ \times SP) - (SQ \times SP)$$

$$= (AQ - SQ)SP$$

$$= (780,000 - 873,000) \$0.006$$

$$= \$ 558 \text{ F}$$



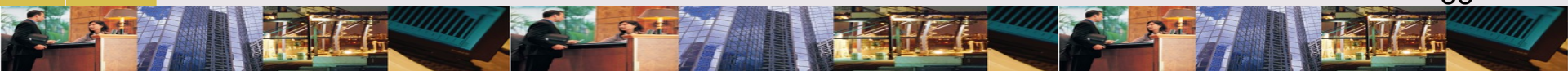
What is the percentage &
source of the favorable
MUV?

**The 10.7% favorable MUV
results from higher quality
corn.**

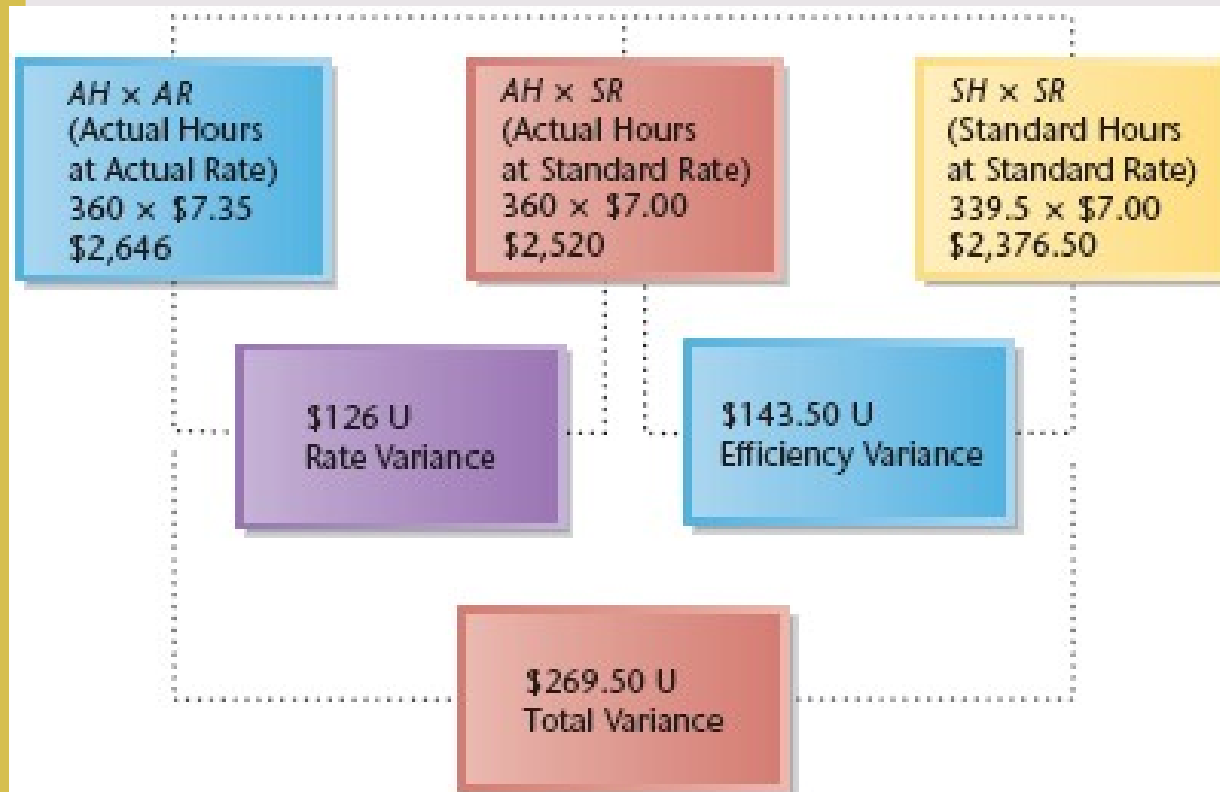


Who is responsible for a materials usage variance?

The Production Manager.



LABOR VARIANCES



Decompose total labor variance into **rate** & **efficiency** variances.

EXHIBIT 9-7

FORMULA: Labor Rate Variance (LRV)

Labor rate variance tells whether a company paid more than expected for labor.

LRV

$$= (AH \times AR) - (AH \times SR)$$

$$= (AR - SR)AH$$

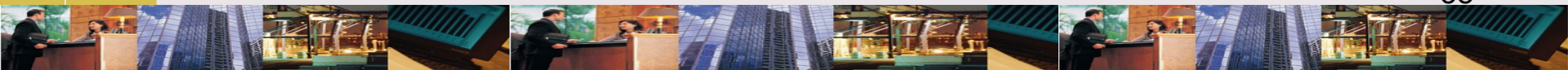
$$= (\$7.35 - \$7.00) 360$$

$$= \$ 126 \text{ U}$$



What is the percentage &
source of the unfavorable
LRV?

**The 5% favorable MUV
results from market forces
& unexpected overtime.**



FORMULA: Labor Efficiency Variance (LEV)

Labor efficiency variance tells whether a company used more labor than expected.

LEV

$$= (AH \times SR) - (SH \times SR)$$

$$= (AH - SH)SR$$

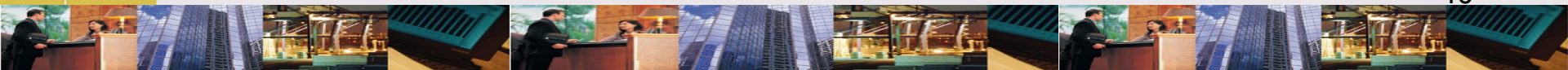
$$= (360 - 339.5) \$7$$

$$= \$ 143.50 \text{ U}$$



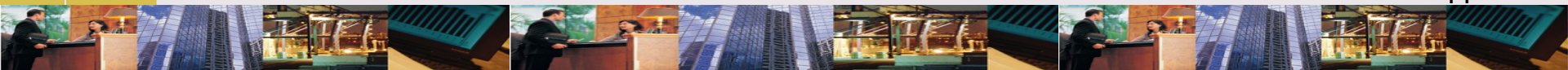
What is the percentage &
source of the unfavorable
LEV?

**The 6% favorable LEV
resulted from machinery
breakdown.**



Who is responsible for a labor efficiency variance?

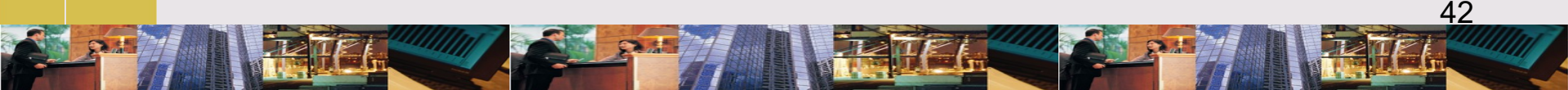
The Production & Maintenance Managers.



LEARNING OBJECTIVE

5

Calculate variable & fixed overhead variances & give their definitions.



VARIABLE OVERHEAD: Background

Variable overhead rate (standard)	\$3.85 per DLH
Actual variable overhead costs	\$1,600
Actual hours worked (machining & inspection)	400
Bags of chips produced	48,500
Hours allowed for production	378.3
Applied variable overhead	\$1,456



FORMULA: Total Variable Overhead Variance

Total overhead variance is the difference between actual and applied variable overhead.

$$\begin{aligned}\text{Total Variable Overhead} \\ &= \text{Actual} - \text{Applied Overhead} \\ &= \$1,600 - \$1,456 \\ &= \$ 144 \text{ U}\end{aligned}$$



VARIABLE OVERHEAD VARIANCE

Decomposes total variable overhead variance into **spending** & **efficiency** variances.

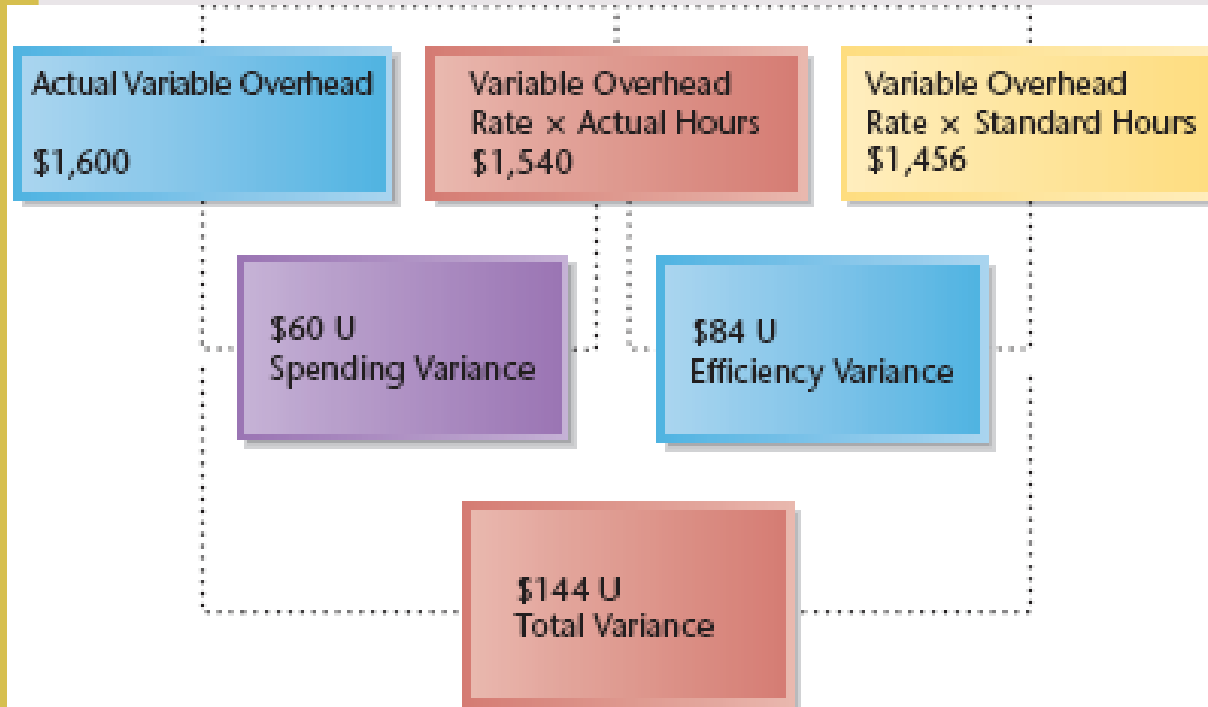


EXHIBIT 9-8

FORMULA: Variable Overhead Spending Variance

Variable overhead spending variance measures aggregate effect of actual variable overhead rate with standard rate.

Spending Variance

$$= (\text{AVOR} \times \text{AH}) - (\text{SVOR} \times \text{AH})$$

$$= (\text{AVOR} - \text{SVOR})\text{AH}$$

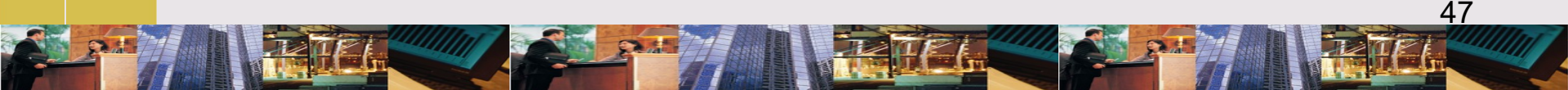
$$= (\$4.00 - \$3.85) 400$$

$$= \$ 60 \text{ U}$$



VARIABLE OVERHEAD SPENDING VARIANCE

Variable overhead spending variance arises because prices change. It includes things such as indirect materials, indirect labor, electricity maintenance, etc. Increase or decrease in these items is beyond control of managers.



FORMULA: Variable Overhead Efficiency Variance

Variable overhead efficiency variance measures change in variable overhead consumption because relies on direct labor.

Efficiency Variance

$$= (\text{AH} - \text{SH})\text{SVOR}$$

$$= (400 - 378.3) \$3.85$$

$$= \$ 84 \text{ U}$$



FIXED OVERHEAD: Background

Budgeted or Planned Items

Budgeted fixed overhead	\$749,970
Practical activity	23,400 DLH
Standard fixed overhead rate	\$32.05

Actual Results

Actual production	2,750,000 bags of chips
Actual fixed overhead cost	\$749,000
Standard hours allowed for actual production	21,450



FORMULA: Applied Fixed Overhead

Applied Fixed Overhead

= SFOR x Standard hours

= \$32.05 x 21,450

= \$ 687,473



FORMULA: Total Fixed Overhead Variance

Total fixed overhead variance is the difference between actual and applied fixed overhead.

$$\begin{aligned}\text{Total Fixed Overhead Variance} &= \text{Actual} - \text{Applied Overhead} \\ &= \$749,000 - \$687,473 \\ &= \$ 61,527 \text{ Underapplied}\end{aligned}$$



FIXED OVERHEAD VARIANCES

Decompose total fixed overhead variance into **spending** & **volume** variances.

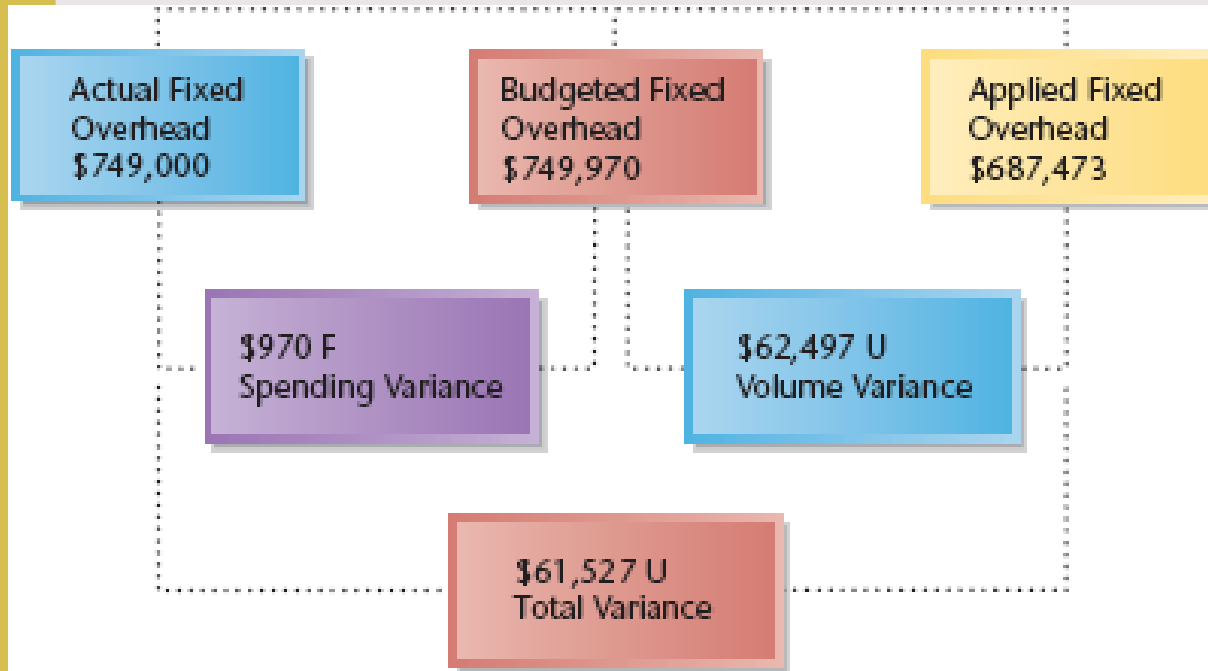
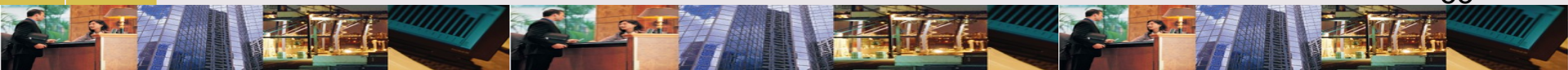


EXHIBIT 9-11

FIXED OVERHEAD SPENDING VARIANCE

Fixed overhead spending variance is the difference between actual and budgeted fixed overhead. It includes things such as salaries, depreciation, taxes, and insurance. Increase or decrease in these items is beyond control of managers.



FORMULA: Fixed Overhead Volume Variance

Fixed overhead volume variance measures the effect of actual output differing from output used to compute predetermined standard fixed overhead rate.

Volume Variance

= Budgeted – Applied fixed overhead

= \$479,970 - \$687,473

= \$62,497 U



LEARNING OBJECTIVE

6

Prepare journal entries
for variances
(*Appendix*).

JOURNAL ENTRIES

Blue Corn must write journal entries to enter information for variances into accounting records. Variances are closed into Cost of Goods Sold.



CHAPTER 9

THE END